

Problem Set 3:

- Assigned: 10/5; Due: Oct. 21
- Same rules as PS2: most problems will be too difficult on a first pass; use any means to solve *and* understand them, and expect to explain what you understand to peers, relatives, strangers on the street, and the instructor.
- Exam 1 (Mon 10/26) is based on PS2 and PS3.
- Topics covered: interference and diffraction, thin films, momentum of light, light in matter (D vs E, Lorentz model, dispersion and absorption), wave packets and group velocity

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1. Unpolarized light and polarizer chains

key ideas: unpolarized light as a mixture, Malus' law averaged, sequential projections, continuum limit

In PS2 you analyzed a beam with fixed E_{0y}/E_{0x} and ϕ . Sunlight and lamp light are not like that: the polarization direction and relative phase wander randomly on a timescale $\tau_c \sim 1/\Delta\nu$ (femtoseconds), far shorter than any detector's response. Such light is *unpolarized*: a mixture of polarizations, not a superposition.

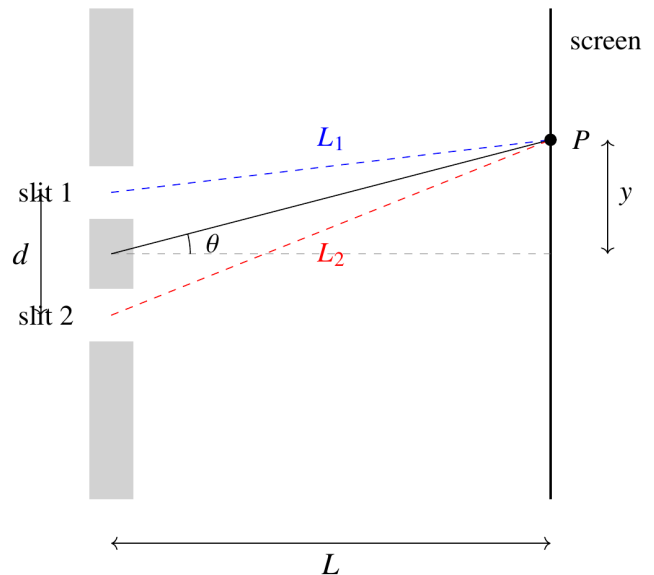
- a. Model unpolarized light as linearly polarized at an angle ψ that is uniformly random over $[0, 2\pi)$. Show that a single polarizer at any angle transmits $I_0/2$, and that no polarizer angle extinguishes the beam.
- b. Two polarizers with transmission axes at 0° and 90° transmit nothing. Insert a third polarizer at angle θ between them and compute the transmitted intensity $I(\theta)$ for unpolarized input I_0 . Show that it is maximal at $\theta = 45^\circ$ and find the maximal value. Write down the field direction after each sheet for $\theta = 45^\circ$: an extra absorber *increases* transmission because it changes the field, not only the intensity.

- c. Replace the middle polarizer by $N - 1$ polarizers at equal angular steps, so that the $N + 1$ sheets (first at 0° , last at 90°) rotate the transmission axis by 90° in N steps of $\pi/2N$. Find I_N/I_0 for polarized input along $\hat{x}$ and show that $I_N/I_0 \rightarrow 1$ as $N \rightarrow \infty$. (Hint: $\cos^{2N}(\pi/2N)$; expand the logarithm.) How many sheets are needed for 90% transmission?
- d. Light reflected at a glancing angle from a horizontal surface (road, water) is partly polarized: take as given that the reflected glare is a mixture of 70% unpolarized light and 30% linearly polarized light with $\vec{E}$ horizontal. Find the transmitted glare intensity through a polarizer as a function of the angle of its axis from vertical, and the angle that minimizes it. What fraction of the glare does the best orientation remove, and what fraction of a fully unpolarized view of the scenery does it cost?

2. Double slit with a glass plate

key ideas: path difference, fringe spacing, finite number of fringes, optical path length, coherence

A Young's double slit has slit separation $d = 0.5$ mm, screen distance $L = 1$ m, wavelength $\lambda = 532$ nm. Point P is at height y on the screen; the path lengths from slits 1 and 2 to P are L_1 and L_2 .



- a. Warm-up. Express $\Delta L = L_2 - L_1$ in terms of d and θ , state the bright/dark conditions on $\Delta L/\lambda$, and for small angles find the fringe positions y_m and spacing Δy (numerically).
- b. Fringes do not continue indefinitely. Find the maximum number of bright and dark fringes on each side of center (correct to ± 1), and the angle θ of the last bright fringe without the small-angle approximation.
- c. A glass plate of thickness t and index $n = 1.6$ covers slit 1. Find the change in optical path difference and the shift of the whole pattern in terms of n , t , L , d . State the direction of the shift (toward or away from the covered slit) and justify it in one sentence without calculation.
- d. Find the minimum t that shifts the pattern by exactly three fringe spacings (numerically). If the fringe shift can be read to 0.1 fringe, to what precision does this measure n for a $t = 25 \mu\text{m}$ plate?
- e. The plate also *delays* the light from slit 1 by $(n - 1)t/c$. For a source with coherence time τ_c , estimate the plate thickness beyond which the fringes wash out at a fixed point near the original center, $y \approx 0$. Evaluate for a laser ($\tau_c \sim 1 \mu\text{s}$) and for a filtered lamp ($\Delta\nu \sim 10^{12}$ Hz).

3. Diffraction: single slit, grating, resolution

key ideas: Huygens sources, phasor sum, grating resolving power, Rayleigh criterion, Bragg reflection

Interference with two sources gave $I = 4I_0 \cos^2(\phi/2)$. Diffraction is the same calculation with many sources. Here you do it for N slits and then let $N \rightarrow \infty$ (a single wide slit) and N large but finite (a grating).

- a. N equally spaced slits (spacing d) each contribute a field of amplitude E_0 with successive phase lag $\delta = kd \sin \theta$. Sum the geometric series $\sum_{j=0}^{N-1} e^{ij\delta}$ and show

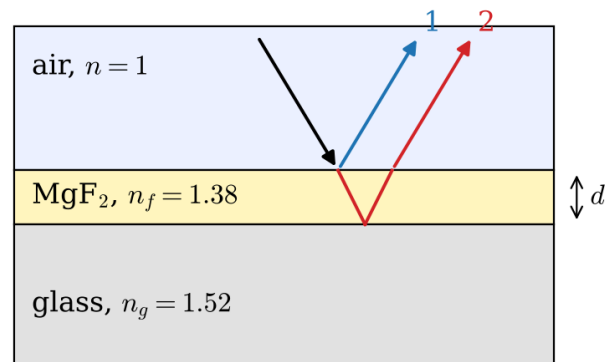
$$I(\theta) = I_0 \frac{\sin^2(N\delta/2)}{\sin^2(\delta/2)}.$$

- Check $N = 2$ against the two-slit result. Where are the principal maxima and what is their height?
- b. Show that the first zero next to a principal maximum is at $\Delta\delta = 2\pi/N$, hence the angular half-width of a grating line is $\Delta\theta \approx \lambda/(Nd \cos \theta)$. Using the Rayleigh criterion (one line's maximum on the other's first zero), derive the resolving power $R = \lambda/\Delta\lambda = Nm$ in order m .
- c. The sodium D lines are at 589.0 and 589.6 nm. How many grating lines must be illuminated to resolve them in first order? In third order? A grating has 300 lines/mm: check that third order exists for this wavelength, and find the illuminated width needed in each case.
- d. Single slit of width a : let $N \rightarrow \infty$ with $Nd = a$ fixed and $E_0 \propto 1/N$. Show that $I(\theta) = I_0(\sin \alpha / \alpha)^2$ with $\alpha = \pi a \sin \theta / \lambda$, and recover the dark-fringe condition $a \sin \theta = m\lambda$, $m \neq 0$. Find the ratio of the first side maximum to the central one.
- e. For a circular aperture the first dark ring is at $\sin \theta = 1.22\lambda/D$ (take as given). At night, the two headlights of an approaching car (1.5 m apart) merge into a single light once the car is farther than about 7 km. Deduce the diameter of the observer's pupil ($\lambda = 550$ nm). In daylight the pupil shrinks to 2 mm: at what distance do the headlights merge then? Apply the same criterion to a 2.4 m telescope mirror: what is the smallest separation it resolves on the Moon (3.8×10^5 km)?
- f. Bragg reflection: the atomic planes of a crystal, spaced d , act as a grating with maxima at $2d \sin \theta = m\lambda$ (take as given). Copper $K\alpha$ X-rays ($\lambda = 0.154$ nm) reflected from NaCl show a first-order peak at $\theta = 15.9^\circ$. Deduce the plane spacing d . Then find the longest wavelength for which *any* Bragg peak exists for this set of planes, and name the part of the spectrum it lies in.

4. Anti-reflection coating

key ideas: reflection phase flips, thin-film interference, unequal reflection amplitudes, index matching, bandwidth

A glass lens ($n_g = 1.52$) is coated with MgF_2 ($n_f = 1.38$) to reduce reflection at normal incidence. Rays 1 and 2 reflect from the air– MgF_2 and MgF_2 –glass interfaces respectively (drawn at a slight angle for clarity). At normal incidence the reflected amplitude at an interface from n_1 into n_2 is $r = (n_1 - n_2)/(n_1 + n_2)$ (take as given; a negative r is the "phase flip").



- a. Identify which of the two reflections flip phase. Write the condition on d for destructive interference between rays 1 and 2 at $\lambda_0 = 550$ nm and find the minimum d .

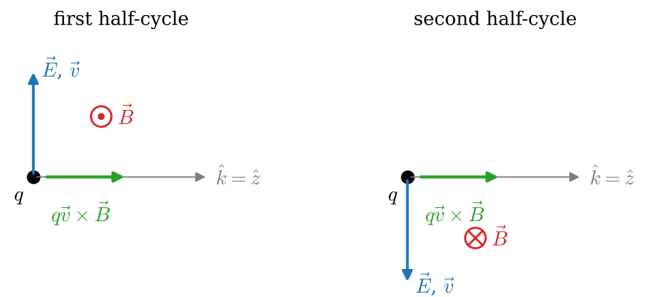
- b. With this d , compute the phase difference $\Delta\phi$ between rays 1 and 2 at 450 and 650 nm. Assuming for now equal amplitudes, the reflected intensity is $\propto |1 + e^{i\Delta\phi}|^2 = 4\cos^2(\Delta\phi/2)$: compute $\cos^2(\Delta\phi/2)$ at the three wavelengths (this is the reflectance relative to its maximum, both reflections in phase). Which colors dominate the reflection from a coated lens in white light?
- c. Now use the actual amplitudes r_1 (air→MgF₂) and r_2 (MgF₂→glass). Neglecting multiple reflections and transmission factors, show the reflected intensity at λ_0 is $\propto (|r_1| - |r_2|)^2$, compare it to the bare-glass reflectance $\propto r_g^2$, and find the coating index n_f that would give exact cancellation.
- d. Bandwidth of the coating. With equal amplitudes the reflectance relative to its maximum is $\cos^2(\Delta\phi/2)$ as in part 2, with $\Delta\phi = 4\pi n_f d/\lambda$ (both reflections flip, so no extra π). Find the two wavelengths at which it has climbed back to one quarter of the maximum, and hence the bandwidth $\Delta\lambda/\lambda_0$ of a single-layer coating. Does it cover the visible band (400–700 nm)?

5. Light carries momentum

key ideas: force on a driven charge, momentum density $\mathbf{g} = \mathbf{S}/c^2$, radiation pressure, solar sails

A laser beam propagates along $+z$, polarized along $\hat{x}$, power $P = 5$ mW, $\lambda = 532$ nm, uniform circular cross-section of radius $w = 1$ mm.

- a. Warm-up. Write $\mathbf{E}$ and $\mathbf{B}$ as a plane wave; from $I = \frac{1}{2}c\epsilon_0 E_0^2$ and the beam area find E_0 numerically; show $u\mathbf{E} = u\mathbf{B}$ and hence $\langle u \rangle = I/c$.



- b. A charge q in an absorbing medium is dragged by $\mathbf{E}$ against friction, so its velocity is in phase with the field, $\mathbf{v} \parallel \mathbf{E}$ at every instant (ignore $\mathbf{B}$ for this step; a free charge would instead have $\mathbf{v}$ a quarter cycle out of phase, and the time-averaged force below would vanish). Show that the magnetic force $q\mathbf{v} \times \mathbf{B}$ points along $\hat{\mathbf{k}}$ in both half-cycles (figure). Show that the charge absorbs energy at rate $qE v$ and momentum at rate $qE v/c$, and conclude that the field carries momentum $\mathbf{p} = U/c$, i.e. momentum density $\mathbf{g} = \mathbf{S}/c^2$.
- c. Radiation pressure on a perfect absorber is I/c and on a perfect mirror $2I/c$: derive both from part 2, compute the force of the beam on a flat absorber and on a flat mirror numerically. The beam is now focused onto a perfectly absorbing dust grain of radius $10 \mu\text{m}$ and density 1000 kg/m^3 (spot smaller than the grain): compare the force with the grain's weight and find the beam power that levitates it.
- d. Sunlight at Earth has $I \approx 1361 \text{ W/m}^2$. Find the pressure on a perfectly reflecting sail and the acceleration of a 1 kg craft with a 100 m^2 sail. How long to reach 1 km/s? Compare the sail's acceleration with the Sun's gravitational acceleration at Earth's orbit ($5.9 \times 10^{-3} \text{ m/s}^2$) and find the sail area per kilogram at which the two balance.

6. Dielectric capacitor: $\mathbf{D}$ vs $\mathbf{E}$ and the Lorentz step response

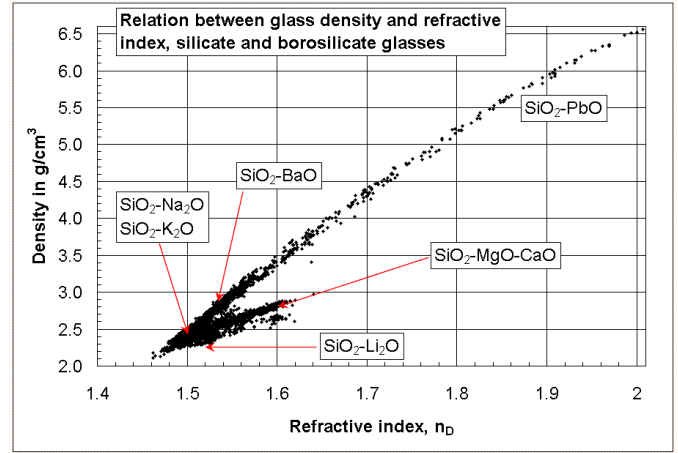
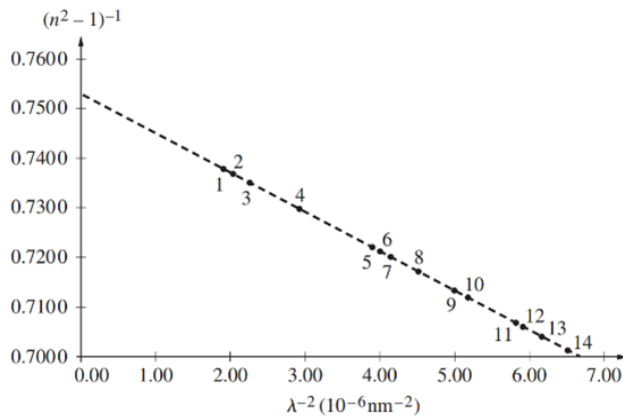
key ideas: Gauss's law in matter, bound charge, energy in a dielectric, step response of a damped oscillator

A thin parallel-plate capacitor (area A , separation $d \ll \sqrt{A}$) is filled with a linear dielectric described by the Lorentz model, $m\ddot{x} + m\gamma\dot{x} + m\omega_0^2 x = qE$, polarization $P = Nqx$. At $t = 0$ free surface charge $\pm\sigma_f A$ is deposited nearly instantaneously on the plates and then held fixed. Write $\omega_p^2 = Nq^2/m\epsilon_0$.

- For t much shorter than every response time of the oscillators ($t \ll 1/\omega_0$ and $t \ll 1/\gamma$) they have not moved ($P = 0$). Use Gauss's law to find $\mathbf{D}$, $\mathbf{E}$, and the initial voltage V_0 .
- For $t \rightarrow \infty$ (long compared with every response time) use the $\omega \rightarrow 0$ limit of the Lorentz model to find P_∞ , hence $\mathbf{E}$, V_∞ , ϵ_r , and the bound surface charge σ_b on the dielectric faces.
- Show that $\mathbf{D}$ is the same in both limits. What about $\mathbf{E}$? Which quantity is "sourced by free charge only"?
- Sketch $E(t)$ and $P(t)$ on the same axes, marking the two limits. Compute $U = \frac{1}{2} \int \mathbf{E} \cdot \mathbf{D} dV$ in both limits; which is larger? Where did the difference go? Show from the Ampère–Maxwell law in matter, $\nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D}$, that $\mathbf{B} = 0$ throughout the relaxation, so no energy enters or leaves through the rim. Then show that the power per unit volume delivered by the field to the oscillators is $\mathbf{E} \cdot \partial_t \mathbf{P}$, and that its time integral equals $U_0 - U_\infty$ plus the potential energy $\frac{1}{2} N m \omega_0^2 x_\infty^2$ stored in the stretched oscillators; conclude that $U_0 - U_\infty$ is exactly the heat produced by the damping γ . (Hint: $\epsilon_0 \partial_t \mathbf{E} = -\partial_t \mathbf{P}$ since $\mathbf{D}$ is constant.)
- Dilute limit: take $\mathbf{E} \approx \mathbf{E}_0 = \sigma_f / \epsilon_0$ as the driving field so the Lorentz equation is the step response of a damped oscillator with $x(0) = \dot{x}(0) = 0$. Solve for $x(t)$ and $\mathbf{E}(t) = \mathbf{E}_0 - Nq x(t) / \epsilon_0$ in the overdamped ($\omega_0 < \gamma/2$) and underdamped ($\omega_0 > \gamma/2$) cases; sketch both, identifying the oscillation frequency $\Omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$ and the $e^{-\gamma t/2}$ envelope.
- In the underdamped case $\mathbf{E}(t)$ overshoots below $\mathbf{E}_\infty$. Find the time of the first minimum of $\mathbf{E}(t)$ and the overshoot $(E_\infty - E_{\min}) / (E_0 - E_\infty)$ in terms of γ / Ω_d . Which of $\mathbf{D}$, $\mathbf{E}$, V overshoot?

7. Lorentz model: dispersion and absorption from data

key ideas: driven oscillator, complex index, Cauchy plot, normal/anomalous dispersion, Beer–Lambert law



An atom in a dielectric responds to $E(t) = \text{Re}[\tilde{E}e^{-i\omega t}]$ as a damped oscillator, $m\ddot{x} + m\gamma\dot{x} + m\omega_0^2 x = qE(t)$; $P = Nqx$ and $n^2 = 1 + P/(\epsilon_0 E)$.

- Solve for $\tilde{x}/\tilde{E}$ and derive $n^2(\omega) = 1 + \frac{Nq^2/m\epsilon_0}{\omega_0^2 - \omega^2 - i\gamma\omega}$.
- With $\gamma = 0$, show that $(n^2 - 1)^{-1}$ is linear in λ^{-2} . From the slope and intercept of the plot (left), extract ω_0 and N (use $q = e$, $m = m_e$). Is ω_0 in the UV, visible, or IR? Compare N to the number density of atoms in glass ($\approx 6.6 \times 10^{28} \text{ m}^{-3}$).
- Sketch $n(\omega)$ for $\gamma = 0$; mark normal and anomalous dispersion, ω_0 , and the width γ once damping is restored.
- Dilute medium: show $n - 1 \propto N$, hence $n - 1 \propto \text{mass density}$. Compare with the plot (right), reading the main ($\text{SiO}_2\text{-PbO}$) branch: is ρ linear in n ? Is it *proportional* to $n - 1$ (does the line extrapolate to $\rho = 0$ at $n = 1$)? Report the slope $d\rho/dn$.
- Restore γ and write $n = n' + i\kappa$. For $E \propto e^{i(kz - \omega t)}$, $k = n\omega/c$, show that intensity decays as $e^{-\alpha z}$ with $\alpha = 2\kappa\omega/c$. At exact resonance with $\gamma \ll \omega_0$ find κ and the absorption length $1/\alpha$; evaluate numerically with N , ω_0 from part 2 and $\gamma/\omega_0 = 10^{-2}$. (Hint: κ is not small here; use $\epsilon' = n'^2 - \kappa^2$, $\epsilon'' = 2n'\kappa$ from lecture rather than expanding.) Compare to the thickness of a window.

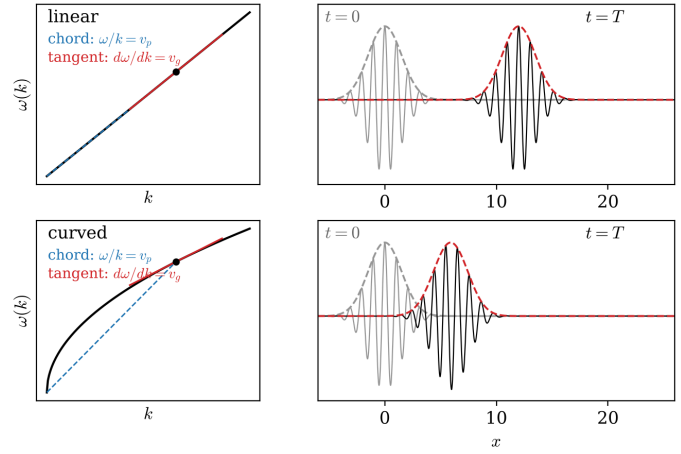
8. Wave packets: where the group velocity comes from

key ideas: superposition, envelope vs carrier, dispersion relation, group velocity, spreading

In lecture, $v_g = d\omega/dk$ came from Taylor-expanding $\omega(k)$. Here you build it from the simplest possible superposition and watch what the curvature of $\omega(k)$ does. Throughout, $\omega(k)$ is the dispersion relation of some medium; its shape is all that matters.

- Add two waves of equal amplitude and nearby wavenumbers, $A \cos(k_1 x - \omega_1 t) + A \cos(k_2 x - \omega_2 t)$ with $k_{1,2} = k_0 \pm \Delta k$ and $\omega_{1,2} = \omega(k_{1,2})$; write $\bar{\omega} = (\omega_1 + \omega_2)/2$ and $\Delta\omega = (\omega_2 - \omega_1)/2$. Show the sum is $2A \cos(\Delta k x - \Delta\omega t) \cos(k_0 x - \bar{\omega} t)$. Identify the carrier and the envelope, give the distance between successive envelope zeros, and show that the carrier crests move at $\bar{\omega}/k_0$ while the envelope moves at $\Delta\omega/\Delta k$.

b. Linear dispersion, $\omega = vk$. Show that both speeds equal v , so the pattern translates rigidly (top row of the figure). Write the sum as a function of $x - vt$ alone.



c. General $\omega(k)$. Let $\Delta k \rightarrow 0$ and show the envelope speed becomes $v_g = d\omega/dk|_{k_0}$, in general different from $v_p = \omega(k_0)/k_0$ (bottom row: the crests outrun the envelope). Apply to deep-water waves, $\omega = \sqrt{gk}$: show $v_g = v_p/2$ and evaluate both for a swell of wavelength 100 m.

d. Apply to the chain of coupled masses from lecture, $\omega(k) = 2\sqrt{\kappa/m} |\sin(ka/2)|$. Find $v_g(k)$; show it equals $a\sqrt{\kappa/m}$ for $ka \ll 1$ and vanishes at $k = \pi/a$. Write out the displacements $u_j(t)$ for $k = \pi/a$ and show that this mode transports nothing: it is a standing pattern in which neighbors move in antiphase.

e. Spreading. Keep three waves, k_0 and $k_0 \pm \Delta k$, with $\omega(k) \approx \omega(k_0) + v_g(k - k_0) + \frac{1}{2}\beta(k - k_0)^2$, $\beta = \omega''(k_0)$. In the frame moving at v_g , show that the two side components acquire a phase $\frac{1}{2}\beta\Delta k^2 t$ relative to the center, so the envelope shape changes appreciably once $\frac{1}{2}|\beta|\Delta k^2 t \sim 1$. For a packet of half-width σ the relevant spread in k is $\Delta k \sim 1/\sigma$: estimate the spreading time $\tau \sim \sigma^2/|\beta|$ for the deep-water swell of part 3 if the packet half-width is five wavelengths.

9. Group velocity in a Lorentz medium; the ionosphere

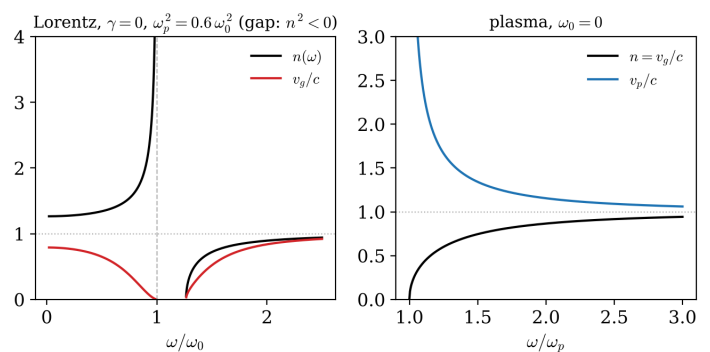
key ideas: phase vs group velocity, $k = n(\omega)\omega/c$, stop band, plasma frequency

A wave packet moves at $v_g = d\omega/dk$, while its crests move at $v_p = \omega/k$. In vacuum both equal c . In a dispersive medium $k = n(\omega)\omega/c$ and the two differ; this problem works out by how much, and what happens where n stops being real.

1. Show that $v_g = \frac{c}{n + \omega dn/d\omega}$. Conclude that

normal dispersion ($dn/d\omega > 0$) gives $v_g < v_p$ and that $v_g < c$ whenever $n > 1$ and $dn/d\omega > 0$.

2. Take the undamped Lorentz model $n^2 = 1 + \omega_p^2/(\omega_0^2 - \omega^2)$ with $\omega_p^2 = Nq^2/m\epsilon_0$. Compute $n + \omega dn/d\omega$ and show that $v_g/c = n \frac{(\omega_0^2 - \omega^2)^2}{(\omega_0^2 - \omega^2)^2 + \omega_p^2 \omega_0^2}$ for $\omega < \omega_0$. Evaluate v_g/c at $\omega \rightarrow 0$ and at $\omega \rightarrow \omega_0^-$; compare with the figure (left).



3. Show that $n^2 < 0$ for $\omega_0 < \omega < \sqrt{\omega_0^2 + \omega_p^2}$. Write the wave in this band as $e^{-z/\ell} e^{-i\omega t}$ and find the penetration depth $\ell(\omega)$; what happens to a wave incident on the medium at such a frequency? Find the width of this band for the glass of problem 7 (use its ω_p and ω_0) in eV.
4. Free electrons: set $\omega_0 = 0$ (the ionosphere, $N \sim 10^{12} \text{ m}^{-3}$). Show $n^2 = 1 - \omega_p^2/\omega^2$, so $n < 1$ and $v_p > c$ for $\omega > \omega_p$, and show $v_p v_g = c^2$ (figure, right). Compute $f_p = \omega_p/2\pi$ numerically.
5. Below f_p the ionosphere reflects (part 3 with $\omega_0 = 0$). Using f_p from part 4, state which of AM radio (1 MHz), FM (100 MHz) and GPS (1.5 GHz) are reflected and which pass. For a signal that passes, the extra travel time through a layer of thickness h is $\Delta t = \int (1/v_g - 1/c) dz$: show that for $\omega \gg \omega_p$ this is $\Delta t \approx h \omega_p^2 / (2c \omega^2)$ and evaluate it for GPS with $h = 300 \text{ km}$. Why does GPS broadcast on two frequencies?